SPU | Syrian Private University
Faculty of Artificial Intelligence Engineering

Project Study and Technical Requirements

Automated Robotic Arm Production Line

Color-based cylinder sorting and reject return in ROS 2

Control and Intelligent Systems

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1. Introduction

This project falls within robotic automation for production lines. It focuses on integrating computer vision, sensors and robotic arms to perform automated sorting. Its aim is to develop a small university simulation model that demonstrates how to distinguish accepted parts from rejected parts and transfer the rejected ones to a separate return path.

The system consists of two parallel conveyors with a four-degree-of-freedom robotic arm positioned between them. Conveyor 1 carries green and red cylinders: green represents accepted parts, while red represents rejected parts for demonstration purposes. A camera identifies the color and estimates the cylinder position and distance from the arm, while a presence sensor confirms its arrival at the pickup position.

Green cylinders continue along conveyor 1. When a red cylinder reaches the pickup area, the conveyor stops. The arm grips the cylinder and transfers it to conveyor 2, which moves in the opposite direction to return it to an area representing repair. The reject conveyor runs when parts are on it and stops after the last part exits.

The framework also includes a display for monitoring part counts, operating times and system status, together with an emergency-stop button. Work during this semester is limited to design and simulation using ROS 2, testing component integration and the operating sequence, and verifying the technical requirements, without building a physical prototype.

2. Literature Review

Paper 1 - Vision-based part localization and conveyor picking

Implementing a Vision-Based ROS Package for Reliable Part Localization and Displacement from Conveyor Belts

Authors: Eber L. Gouveia, John G. Lyons, Declan M. Devine

Published in: Journal of Manufacturing and Materials Processing, Vol. 8, Article 218. **Publisher:** MDPI, Switzerland. **Publication year:** 2024

Affiliation and country: Technological University of the Shannon, Ireland.

Brief content: The paper presents a flexible ROS-based 3D-vision framework for locating parts and picking them from a conveyor, with reconfiguration to accommodate changes in their positions and orientations.

Paper 2 - Robot and conveyor layout planning for box sorting

Budget-optimal Multi-robot Layout Design for Box Sorting

Authors: Peiyu Zeng, Yijiang Huang, Simon Huber, Stelian Coros

Published in: arXiv:2412.11281 (preprint). **Publisher:** arXiv. **Publication year:** 2024; revised 7 March 2025

Affiliation and country: Swiss Federal Institute of Technology in Zurich (ETH Zurich), Switzerland.

Brief content: The paper studies robot and conveyor placement for minimum-cost box transport, using a reachability graph and mathematical optimization while verifying motion feasibility.

Paper 3 - Automated textile sorting using a robot and digital twin

Autonomous Textile Sorting Facility and Digital Twin Utilizing an AI-Reinforced Collaborative Robot

Authors: Torbjørn Seim Halvorsen, Ilya Tyapin, Ajit Jha

Published in: Electronics, Vol. 14, Article 2706. **Publisher:** MDPI, Switzerland. **Publication year:** 2025

Affiliation and country: University of Agder, Norway.

Brief content: The paper presents a textile-sorting system combining computer vision, recognition, tracking and robotic handling, with ROS 2, a digital twin and motion simulation.

Paper 4 - Color recognition and arm-conveyor synchronization

Implementation of HSV Algorithm for Speed Synchronization of Robot Arm and Conveyor

Authors: Noval Lilansa, Fitria Suryatini, Dede Moch. Dinar Wahidin

Published in: ELKOMIKA, Vol. 13, No. 2, pp. 170-184. **Publisher:** Department of Electrical Engineering, Institut Teknologi Nasional Bandung, Indonesia. **Publication year:** 2025

Affiliation and country: Politeknik Manufaktur Bandung, Indonesia.

Brief content: The paper uses HSV and a camera to identify object color and position, estimate speed, and coordinate arm motion to pick objects from a moving conveyor.

2. Literature Review - Comprehensive Comparison

Comparison element	Study 1: ROS-based part localization - 2024	Study 2: Robot and conveyor layout - 2024/2025	Study 3: Textile sorting and digital twin - 2025	Study 4: HSV arm-conveyor synchronization - 2025
Main objective	Locate, pick and transfer parts using vision	Select robot and conveyor positions for transport at minimum equipment cost	Recognize and automatically sort textiles with a digital system representation	Recognize colored objects and estimate position and speed for picking in motion
Arm degrees of freedom	6 DOF	6 DOF in the tested arm models	6 DOF	4 DOF, according to the specification of the robot model used
Gripper type	GIMATIC PB-0013 pneumatic gripper; pressure closes it and a spring opens it	Not reported	Custom pneumatic gripper combining a scoop and claw	Not clearly specified
Camera or vision device	Photoneo PhoXi 3D Scanner S	No physical camera documented; geometric dimensions are model inputs	Intel RealSense D455 color-and-depth camera (RGB-D)	Logitech C270 2D color webcam
Vision function	Determine part position, orientation, shape and dimensions	No physical vision implemented; supplied geometric information is used	Recognize the object class, track it and determine its 3D position	Detect color, locate the object center and coordinates, and estimate speed
Part-presence sensor	Two retroreflective photoelectric sensors	Not reported	No separate presence sensor reported; tracking relies on vision	No separate presence sensor reported; measurements rely on the camera
Sorting or routing criterion	Shape, dimensions and matching to a reference shape	Deliver boxes to specified destinations	Textile color classes: light, dark and multicolor	Object color
Recognition algorithm or processing method	DBSCAN for point clustering, PCA for orientation estimation, and shape matching	Reachability graph and MILP for layout optimization; no color-detection algorithm	YOLOv8m for detection and classification; BoT-SORT for tracking	HSV color segmentation and object-center identification
Conveyors used	One straight conveyor	Conveyor networks with different configurations	One conveyor	One conveyor
Conveyor state during picking	Stops for inspection and picking	Transport operations are studied in kinematic simulation	Stops for picking under system coordination, then resumes	Keeps moving during picking
Software and simulation	ROS, MoveIt and RViz; digital model and motion visualization	Unnamed kinematic simulator; Gurobi solves the optimization problem	ROS 2, MoveIt 2, Gazebo and RViz	OpenCV and parallel task processing; physical prototype, with no documented ROS or simulator use
Benefit to our project	Determine pickup coordinates, detect arrival and coordinate conveyor stopping	Choose the arm position, conveyor spacing and transfer points within reach	Integrate the camera, arm and conveyor, and organize simulation and the operating cycle	Distinguish red from green and locate the cylinder center using vision
Limits of relevance to our project	Uses 3D equipment and shape/size sorting, with one conveyor	Does not address color detection or a vision-based rejection cycle	Handles textiles requiring a special gripper and uses one conveyor	Moving-conveyor picking needs more precise timing; our project stops the conveyor to pick red cylinders

"Not reported" means that the paper does not clarify the element; it does not necessarily mean the element is absent.

Benefit of Each Paper to Our Project

Paper 1: We will draw on it to convert vision data into suitable arm coordinates and coordinate arrival detection, conveyor stopping and picking.
Paper 2: We will draw on it to choose the arm position, conveyor spacing, and pickup and placement points within reach.
Paper 3: We will draw on it to integrate the camera, arm and conveyor using ROS 2, organize the simulation, and coordinate stopping, picking and restarting.
Paper 4: We will draw on it to distinguish red from green and locate the cylinder center and position using vision in a small educational model.

3. Project Objectives

Main Objective

> Design and simulate an automated sorting system using ROS 2, consisting of two parallel conveyors and a four-degree-of-freedom robotic arm, to sort cylinders by color. Green cylinders continue along their path, while red cylinders, representing rejected parts, are transferred to an oppositely moving conveyor that returns them to the repair area.

Secondary Objectives

1. **Color recognition:** Distinguish red and green cylinders using the camera.
2. **Part localization:** Estimate the cylinder distance and coordinates relative to the arm for correct picking.
3. **Motion coordination:** Stop conveyor 1 when a red cylinder reaches the pickup area, then resume after transfer. A green cylinder does not request a conveyor stop.
4. **Non-damaging transfer:** Pick up the cylinder and place it on conveyor 2 without deformation or dropping, using enough grip pressure to retain it without harmful over-squeezing, while minimizing operation time and maintaining stability.
5. **Reduce unnecessary operation:** Run the reject conveyor when parts are on it and stop it after the last part exits.
6. **Monitoring:** Display accepted and rejected part counts, conveyor and arm operating times, system status and faults.
7. **Safety:** Provide an emergency-stop button that stops system motion without dropping a held part.

4. Technical Requirements

Preliminary Technical Specifications - Cylinder Sorting and Rejection Return

Project Objectives and Scope	
Main objective	Sort red and green cylinders. Green cylinders continue on conveyor 1; red cylinders represent rejected parts and are transferred by the arm to conveyor 2, which returns them in the opposite direction toward the repair area.
Secondary objectives	Detect rejected parts, measure their distance from the arm, coordinate transfer, hold and transfer the cylinder without deformation or damage using sufficient grip pressure without harmful over-squeezing, and minimize operation time.
Scope	A small university model, implemented at this stage through ROS 2 simulation.
Physical and Kinematic Constraints	
System arrangement	Two parallel belt conveyors, with one four-degree-of-freedom robotic arm positioned between them.
Motion and reach	The arm shall pick and transfer the cylinder within a working distance of less than 60 cm from its base.

4. Technical Requirements

Payload and Performance Requirements	
Part mass	The arm shall lift and transfer a cylinder with a mass of 50 g, which is below 100 g.
Conveyor load	The maximum total load on a conveyor shall be 4 kg.
Actuation and Energy Requirements	
Conveyor speed	Initial operating speed: 5 cm/s, adjustable to suit stable detection, stopping and picking in the small model.
Reject-conveyor operation	Conveyor 2 shall run when a rejected part is on it and stop after the last part exits, reducing unnecessary operation and energy consumption.
Operation time	Complete the transfer in the shortest practical time while maintaining part stability and integrity.
Sensor and Controller Requirements	
Camera function	Use a camera to distinguish red from green and measure the cylinder distance from the arm.
Presence and conveyor stopping	Use a presence sensor to confirm arrival at the pickup position. Stop conveyor 1 when a red cylinder arrives; green cylinders shall not request a stop. Resume after transfer and arm clearance.

4. Technical Requirements

Communication and Functional Interface Requirements	
Runtime counters	Provide a display showing total experiment time and accumulated active operating time for each conveyor and the arm separately, from the start of the experiment.
Part counters	Display the number of red rejected parts, green accepted parts and the total. Count each part once.
System status	Display whether both conveyors and the arm are running or stopped, with alerts for current faults.
Safety and Reliability Requirements	
Emergency stop	Provide a clearly visible on-screen emergency-stop button. Pressing it shall immediately stop both conveyors and arm motion without releasing a held part. Motion shall resume only after a new start command following fault resolution.